

Analyzing and Forecasting River Morphological Evolution Using Machine Learning and Spatiotemporal Neural Models: A Comprehensive Case Study on the Padma River, Bangladesh

Abstract—Rivers are dynamic geomorphological systems experiencing continuous transformation through erosion and sediment transport. In Bangladesh, these processes displace 50,000–200,000 people annually, causing severe losses along the Padma River. Traditional monitoring via field surveys and manual remote sensing proves costly, spatially limited, and unsuitable for multi-year prediction. This paper presents a comprehensive framework for forecasting river morphological evolution from freely available satellite data using spatiotemporal deep learning. A 38-year time series (1987–2025) of binary water masks was constructed from Landsat (5 TM, 7 ETM+, 8 OLI) and Sentinel (Sentinel-2 MSI, Sentinel-1 SAR) imagery via Google Earth Engine. Water extraction used Modified Normalised Difference Water Index (MNDWI). The proposed Bidirectional ConvLSTM gap-filling model achieved IoU = 0.7366, outperforming classical methods. Five spatiotemporal architectures (ConvLSTM, U-Net+LSTM, Attention U-Net+ConvLSTM, Swin Transformer, ViT-based model) were evaluated across yearly, quarterly, and bi-monthly resolutions. Hybrid CNN-LSTM models consistently outperformed pure transformers. Attention U-Net+ConvLSTM achieved best yearly performance (IoU = 0.7005); U-Net+LSTM led bi-monthly prediction (IoU = 0.7791). Statistical analysis quantified mean annual migration of 255.4 m yr^{-1} with 1998 extreme of $1,485.5 \text{ m yr}^{-1}$. Long-term forecasting (2026–2040) projected 290 km^2 expansion with spatially explicit risk maps. Results demonstrate freely available satellite imagery with deep learning provides practical, scalable framework for riverbank hazard monitoring and disaster preparedness.

Index Terms—River morphology; Riverbank erosion forecasting; Spatiotemporal neural networks; ConvLSTM; Attention mechanisms; U-Net; Swin Transformer; Vision Transformer; Satellite remote sensing; MNDWI; Google Earth Engine; Padma River; Bangladesh; Time series prediction; Disaster management; Gap-filling; Deep learning; Flood forecasting

I. INTRODUCTION

Rivers constitute among the most morphologically dynamic features characterizing Earth's terrestrial surface. Continuous geomorphological reshaping transpires through interconnected processes of fluvial erosion, suspended and bedload sediment transport, and depositional accretion across extraordinarily wide-ranging spatial and temporal scales: from diurnal fluctuations manifested during individual flood events to decadal shifts in channel position, planform geometry, cross-sectional morphology, and longitudinal profile characteristics. Comprehensive understanding and reliable predictive capability regarding these morphodynamic transformations proves fun-

damental to effective integrated water resource management, safeguarding of human settlements and infrastructure, maintenance of agricultural productivity in riparian zones, and preservation of critical riparian and aquatic ecosystem integrity and biodiversity [1].

Bangladesh occupies a singularly precarious geomorphological and hydrological position within this global context. The nation forms the terminal deltaic sink of the integrated Ganges-Brahmaputra-Meghna river system—representing Earth's most extensive active delta complex—and its extensive river network exhibits characteristically exceptional suspended sediment loads (exceeding 1 billion tonnes annually), intense seasonal flooding driven predominantly by South Asian monsoon meteorological forcing, and documentation of some of the highest lateral channel migration rates recorded anywhere globally, frequently exceeding several hundred metres per individual monsoon season and occasionally reaching kilometre-scale adjustments [2], [3]. These extreme morphodynamic processes generate persistent and severe riverbank erosion phenomena that systematically displace between 50,000 and 200,000 individuals annually and inflict devastating cumulative economic losses through wholesale destruction of agricultural land, transportation infrastructure (roads, bridges, ferries), flood protection embankments, educational facilities, health-care infrastructure, and the fundamental social fabric binding riverine communities [4]. The cascading socioeconomic consequences propagate far beyond immediate physical displacement to encompass long-term food insecurity among displaced populations, systematic disruption of educational access for affected children, dissolution of established social networks and community structures, and profoundly constrained livelihood trajectories and economic opportunities, with demonstrably disproportionate burdens borne by the most economically vulnerable households lacking resources for adaptation or relocation [3], [5], [6].

Despite more than five decades of continuously accumulated satellite remote sensing imagery archives now being comprehensively available through multiple international space agencies and data repositories, these invaluable spatiotemporal datasets remain substantially under-exploited for predictive morphological modelling applications beyond retrospective change detection. The dominant research paradigm character-

izing existing peer-reviewed literature emphasizes retrospective change detection methodologies: quantitative comparison of discrete temporal snapshots to characterize and quantify morphological transformations that have already occurred, rather than developing prospective forecasting capabilities to predict transformations that will occur. Classical remote sensing analytical approaches inherently overlook the continuous temporal evolution trajectories and complex spatial dependencies intrinsically characterizing river morphodynamic processes. Recent transformative breakthroughs in deep spatiotemporal learning architectures, particularly computational frameworks jointly exploiting convolutional spatial feature extraction and recurrent temporal memory mechanisms, have demonstrated unprecedented success in learning extraordinarily complex sequential patterns from high-dimensional image sequence data across diverse application domains [7], [8]. However, systematic application of these powerful architectures specifically to multi-year river morphology spatial forecasting remains comparatively sparse within published literature, and no comprehensive comparative evaluation rigorously assessing multiple competing architectures for this specific forecasting task under controlled experimental conditions has previously been reported.

II. LITERATURE REVIEW

Riverbank erosion monitoring has evolved substantially with satellite remote sensing technology. Early work relied on aerial photography and manual channel boundary digitization—labor-intensive approaches limited to small spatial extents. The Landsat mission series, spanning five decades of continuous observation, revolutionized the field by enabling systematic, large-area, long-term monitoring at unprecedented scales.

A. Optical Satellite-Based Monitoring

Binh et al. [9] developed the Sentinel-based River Bank Erosion Detection (SRBED) framework for automated riverbank extraction from Landsat time series, demonstrating substantial potential for large-scale morphological change detection. Langhorst et al. [10] extended this globally, constructing algorithms detecting lateral bank movement across diverse river systems and identifying erosion hotspots at continental scales. These contributions established critical baseline erosion rates, revealing extraordinary severity of riverbank retreat in South and Southeast Asia. Langat et al. [11] and Islam and Haque [12] demonstrated multi-temporal remote sensing with GIS for quantifying channel migration trends in Bangladesh. Raju et al. [13] analyzed erosion–accretion dynamics, highlighting heterogeneous distribution of geomorphically active zones along river reaches where erosion alternates spatially with stable and accretionary zones.

B. Synthetic Aperture Radar Applications

Optical remote sensing in tropical environments suffers from persistent cloud cover, particularly during monsoon seasons when erosion peaks. Synthetic Aperture Radar (SAR) provides cloud-independent all-weather observation. Freihardt

et al. [2] used Sentinel-1 SAR time series in Google Earth Engine to map riverbank erosion in Bangladesh, documenting rates exceeding 100 m/year and validating SAR water masking using VV backscatter threshold of -16 dB. DeVries et al. [24] demonstrated that combining Sentinel-1 and Landsat in Google Earth Engine enables rapid flood mapping scalable to continental extents. Sazib et al. [27] extended SAR-based mapping across the Ganga-Brahmaputra basin, linking inundation extents with agricultural exposure and population impact assessments.

C. Regional Case Studies

Several studies focused on specific river systems. Tha et al. [14] evaluated riverbank erosion hotspots along Cambodia's Mekong River, integrating physical erosion analysis with vulnerability assessments connecting geophysical processes with socioeconomic consequences. Saha [15] conducted multi-sensor analysis at the Padma-Jamuna confluence, documenting complex erosion, deposition, and island formation patterns from junction dynamics. Haque [16] studied multi-decadal Jamuna River changes, quantifying agricultural land losses and settlement displacement. Ritu et al. [17] modeled Padma River bank shifting; however, 10-year intervals between acquisitions limited prediction accuracy and seasonal sensitivity. Muzahid et al. [18] quantified erosion and accretion along the Gorai River using multi-temporal Landsat classification.

D. Advanced Segmentation and Risk Assessment

Rafat et al. [19] applied the Segment Anything Model (SAM) to satellite imagery for mapping erosion-prone riverbanks and lost villages in Bangladesh, demonstrating that modern computer vision foundation models enhance erosion mapping accuracy. However, SAM focused on historical mapping without temporal sequence modeling or predictive capability. Ren [20] developed large-scale erosion risk assessment combining remote sensing with topographic, hydrological, and land-cover variables. A fundamental limitation shared across these studies is absence of multi-year predictive modeling grounded in spatiotemporal deep learning.

E. Socioeconomic Dimensions

Human dimensions of riverbank erosion provide strong motivation for improved predictive tools. Islam et al. [4] documented displacement patterns and economic losses, showing erosion-induced migration drives rapid urbanization and informal settlement growth. Das et al. [5] documented psychological trauma from displacement and livelihood reconstruction impairment, emphasizing impacts extend beyond economic losses to encompass social cohesion and community resilience. Alam et al. [3] and Sarkar et al. [6] examined livelihood vulnerability among riparian households along the Padma and in northern Bangladesh, revealing poorest households face disproportionate exposure due to limited adaptive capacity and land-based livelihood dependence.

III. RELATED WORK AND RESEARCH GAPS

A. Deep Learning for Water Body Segmentation

Recent deep learning advances have supplanted traditional threshold-based water extraction methods. Yuan et al. [21] demonstrated CNNs achieve superior accuracy over NDWI/MNDWI baselines across scenes with mixed land cover, shadows, and variable turbidity. Li et al. [22] proposed attention-enhanced multi-scale residual U-Net for water segmentation from Sentinel-2 imagery, showing attention mechanisms effectively discriminate water from spectrally similar classes like shadows and saturated soils. Ghosh et al. [23] advanced multi-scale fusion within U-Net architectures, achieving enhanced detection and improved edge definition for rivers of varying dimensions. These architectures process individual images without temporal awareness and cannot forecast morphological change.

B. Cloud-Based Processing Infrastructure

Google Earth Engine (GEE) has democratized large-scale satellite processing, enabling operations on multi-decadal archives without local download. DeVries et al. [24] established platform suitability for operational flood monitoring. Clement et al. [25] applied GEE to ephemeral river flooding, and Tripathy and Mallick [26] extended it to large-scale Mekong River monitoring, demonstrating continental scalability.

C. Recurrent and Hybrid Networks for Hydrological Forecasting

Fan et al. [7] applied LSTM to river discharge forecasting, demonstrating temporal relationships in hydrological data can be learned to generate superior predictions versus autoregressive models. Le et al. [28] and Xiang et al. [29] validated hybrid CNN-LSTM frameworks for river flow prediction, showing complementary gains from combining spatial feature extraction with temporal memory. Horvath et al. [30] highlighted importance of variable selection and temporal lag configuration for LSTM water level prediction on the Tisza River. Li et al. [22] demonstrated interpretability value of attention mechanisms in CNN-LSTM models for streamflow forecasting across the Qinghai-Tibet Plateau.

D. ConvLSTM and Spatiotemporal Architectures

ConvLSTM architectures replace matrix multiplications in recurrent gates with convolutions, preserving spatial locality during temporal processing. Dehghani et al. [8] provided systematic comparative evaluation of LSTM, CNN, and ConvLSTM for spatiotemporal forecasting, finding ConvLSTM superior for tasks with strong spatial dependence but noting higher computational cost. Lemenkova [1] applied deep learning including convolutional-recurrent architectures to track flood dynamics in the Ganges Delta, demonstrating applicability to water extent data in the same geographical context. For transformer-based video understanding, Liu et al. [31] proposed Video Swin Transformer with hierarchical window-based self-attention, while Dosovitskiy et al. [32]

established the ViT framework and Arnab et al. [33] extended it to video (ViViT), providing architectural foundations for spatiotemporal transformer models evaluated here.

E. Identified Research Gaps

Comprehensive literature review reveals four critical gaps motivating this investigation:

First, the vast majority of existing research focuses on historical change detection rather than forecasting future morphological states.

Second, while deep learning has been applied to water body segmentation and streamflow prediction, its application to multi-year spatial river morphology forecasting remains limited.

Third, no systematic cross-architecture comparison for this specific forecasting task has been published under controlled experimental conditions.

Fourth, physical geomorphological and socioeconomic perspectives are rarely integrated in unified predictive frameworks suitable for operational disaster management.

F. Novelty and Contributions

This investigation addresses these gaps through: (i) sequence-based learning treating river evolution as continuous spatiotemporal processes using multi-year sliding windows to capture decadal trends; (ii) first systematic five-architecture comparison spanning CNN-LSTM hybrids and transformers under identical experimental conditions; (iii) extension of prediction horizons to 15-year timescales relevant for infrastructure planning and disaster preparedness; (iv) attention mechanisms enabling automatic identification of morphologically active zones; and (v) foundational framework enabling future integration with socioeconomic risk mapping for operational early warning systems.

IV. STUDY AREA AND DATASET CONSTRUCTION

A. Study Area: Padma River Mid-Course Reach

The designated study area (Fig. 1) encompasses an extensive mid-course longitudinal section of the Padma River, incorporating its active channel corridor, confluent or immediately proximal floodplain zones, and prominent mid-channel island and bar geomorphological complexes. The spatial region of interest was strategically delineated to comprehensively capture dominant morphological evolution patterns and characteristic channel migration behaviours within a computationally tractable spatial domain suitable for intensive deep learning model training.

The Padma River forms through confluence of the Ganges (locally termed Padma upstream of junction) and Jamuna Rivers at Goalundo Ghat and flows approximately 120 km in southeastward direction before ultimately joining the Meghna River to discharge into the Bay of Bengal through multiple tributary channels. The river annually transports multiple millions of tonnes of Himalayan-derived suspended and bed-load sediment and exhibits characteristically highly dynamic morphodynamic behaviour including intense lateral channel

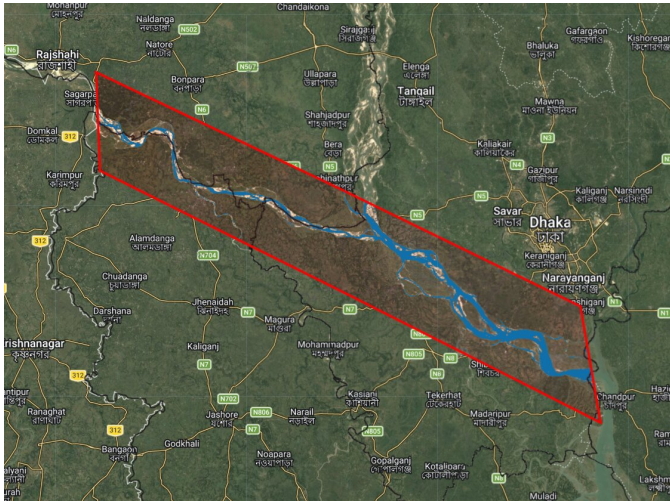


Fig. 1. Study area delineation along mid-course section of Padma River, Bangladesh.

migration, extreme bank erosion rates frequently exceeding 500 m during individual monsoon seasons, and regular mid-channel bar development, modification, and dissolution cycles. The discharge regime exhibits pronounced monsoonal characteristics: peak flow occurs during July–September monsoon season, during which the river maintains a complex braided planform geometry with rapidly changing channel configurations and extensive mid-channel island features. During dry season months (January–April), flow contracts substantially to form much narrower single-thread or limited multi-thread channel configurations with extensive exposed bar surfaces.

The active river corridor sustains dense riparian population concentrations on both banklines, with the river serving multiple critical socioeconomic functions encompassing passenger and cargo transportation, agricultural irrigation water supply, artisanal fisheries supporting local livelihoods, and potable water supply for riparian communities. Despite this fundamental socioeconomic importance, characteristically extreme erosion rates have generated significant cumulative losses of productive agricultural land, systematic large-scale household displacement events, and wholesale destruction of critical transportation infrastructure (roads, bridges, ferry terminals), educational facilities, and healthcare infrastructure [3].

B. Comprehensive Satellite Data Acquisition Protocol

Satellite imagery acquisition was strategically organized into two temporally distinct operational phases based on systematic sensor availability patterns and mission operational timelines.

Phase I: Landsat Mission Data (1987–2014). Multispectral optical imagery was sourced exclusively from United States Geological Survey (USGS) Landsat Collection 2 Level-2 atmospherically-corrected surface reflectance product archives accessed through Google Earth Engine Application Programming Interface (API). Three successive sensor platforms were employed sequentially follow-

ing their respective operational timelines: Landsat 5 Thematic Mapper (TM) sensor (GEE collection identifier: LANDSAT/LT05/C02/T1_L2) operational prior to 1999; Landsat 7 Enhanced Thematic Mapper Plus (ETM+) sensor (LANDSAT/LE07/C02/T1_L2) operational from 1999 onward; and Landsat 8 Operational Land Imager (OLI) sensor (LANDSAT/LC08/C02/T1_L2) operational from 2013 onward. All three sensors provide systematically consistent 60 m spatial resolution imagery (resampled from native 30 m for consistency with SWIR bands) with 16-day temporal revisit frequency. Scene-level cloud filtering applied maximum permissible cloud fraction threshold of 50% via CLOUD_COVER image-level metadata attribute. Pixel-level cloud and cloud-shadow masking protocols utilized QA_PIXEL quality assessment band via bitwise logical operations, specifically targeting bit 3 (cloud shadow detection) and bit 4 (cloud detection) binary flags. Landsat Collection 2 Level-2 products provide surface reflectance as scaled integers; conversion to physical surface reflectance values employed standardized scaling relationship:

$$\rho_{\lambda} = 0.0000275 \times DN_{\lambda} - 0.2 \quad (1)$$

where ρ_{λ} represents surface reflectance for spectral band λ and DN_{λ} represents digital number value. Spectral band selections for MNDWI computation varied systematically by sensor platform following band designation conventions: Band 2 (green, 520–600 nm) and Band 5 (SWIR1, 1550–1750 nm) for Landsat 5 and Landsat 7 platforms; Band 3 (green) and Band 6 (SWIR1) for Landsat 8 platform. Temporal composite imagery was systematically generated for three distinct temporal aggregation schemes: yearly (single annual composite per calendar year), quarterly (four three-month period composites per calendar year: Q1 January–March, Q2 April–June, Q3 July–September, Q4 October–December), and bi-monthly (six two-month period composites per calendar year: P1 January–February, P2 March–April, P3 May–June, P4 July–August, P5 September–October, P6 November–December).

Phase II: Sentinel Mission Data (2015–2025). Sentinel-2 MultiSpectral Instrument (MSI) optical data were systematically accessed via Google Earth Engine collection identifier COPERNICUS/S2_SR_HARMONIZED, representing harmonized Level-2A bottom-of-atmosphere reflectance products. Scene-level filtering applied maximum cloud cover threshold of 50% based on CLOUDY_PIXEL_PERCENTAGE metadata attribute, with scenes subsequently sorted in ascending order of cloud fraction to prioritize least contaminated observations. To prevent computational timeout errors during large-scale export operations, maximum number of Sentinel-2 scenes utilized per temporal composite was algorithmically capped at 25 images. Pixel-level cloud masking protocols utilized QA60 quality assessment band targeting bit 10 (opaque cloud detection) and bit 11 (cirrus cloud detection) binary flags. Sentinel-2 MSI products provide surface reflectance as physical values requiring no additional radiometric scaling transformations.

To systematically address persistent data gap patterns arising from sustained cloud cover contamination particularly

pronounced during monsoon seasons, Sentinel-1 Synthetic Aperture Radar (SAR) C-band data were strategically incorporated as supplementary fallback data source activated when Sentinel-2 optical imagery proved unavailable. Sentinel-1 data were systematically accessed from Google Earth Engine collection COPENNICUS/S1_GRD representing Ground Range Detected (GRD) products acquired in Interferometric Wide (IW) swath mode with VV and VH dual polarization configuration in descending orbital pass geometry. Scene selection was sorted by acquisition date in descending temporal order and algorithmically limited to maximum of 20 images per temporal composite to manage computational processing load. The SAR-based fallback protocol was activated exclusively when zero Sentinel-2 scenes successfully passed cloud filtering criteria for specified temporal window. Under these fallback conditions, water mask derivation employed VV polarization backscatter coefficient with empirically-determined fixed threshold of -16 dB following established protocols [2].

All systematically generated water masks were exported as georeferenced GeoTIFF raster files in geographic coordinate system EPSG:4326 (WGS84 datum). An export spatial scale of 60 m was consistently applied as primary spatial resolution across all platforms to maintain systematic consistency, with automated fallback to full bounding-box spatial extent in exceptional cases where computational resource limitations were exceeded.

C. Comprehensive Image Availability and Quality Assessment

Table I provides systematic summary of image availability characteristics and quality assessment metrics evaluated across four distinct temporal resolution aggregation schemes. The yearly dataset achieved complete temporal coverage (38 observations spanning 1988–2025, zero missing observations), thereby providing most stable and reliable foundation for long-term temporal trend analysis applications. The monthly resolution dataset exhibited most severe data loss patterns, with 50 completely missing temporal observations and 122 partial or low-quality degraded observations, thereby precluding its systematic utilization for primary modelling experiments. Bi-monthly and quarterly datasets demonstrated progressively improving completeness characteristics, with quarterly temporal resolution providing optimal balance between temporal granularity and systematic data reliability (only 3 completely missing observations across 38-year period).

Subsequent comprehensive manual inspection procedures complemented automated filtering protocols to systematically identify partially missing images exhibiting localized spatial gaps or degraded regions not adequately captured by cloud-fraction threshold criteria alone (Fig. 2). These quality impairment patterns proved most prevalent within bi-monthly dataset during July–August temporal periods (peak monsoon season) when persistent cloud cover systematically impedes optical observation capabilities.

TABLE I
SYSTEMATIC SUMMARY OF IMAGE AVAILABILITY AND QUALITY
ASSESSMENT ACROSS MULTIPLE TEMPORAL RESOLUTION SCHEMES

Temporal Resolution	Usable Images	Missing Images	Partial/Low-Quality Images
Monthly	284	50	122
Bi-monthly	156	16	56
Quarterly	125	3	24
Yearly	38	0	0

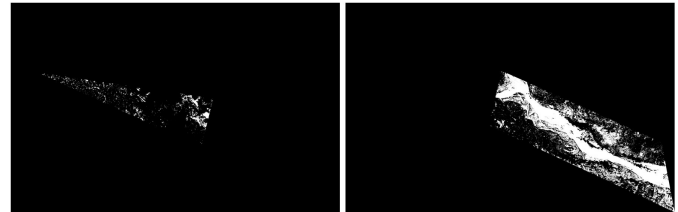


Fig. 2. Partially Missing Samples

D. Water Body Detection via Modified Normalized Difference Water Index

Water body extraction procedures employed Modified Normalised Difference Water Index (MNDWI) spectral index calculations [2], [14], which systematically exploit contrasting spectral reflectance behaviour characteristics of water surfaces (characteristically high reflectance in green visible wavelengths, strong absorption in shortwave infrared wavelengths) relative to terrestrial surfaces including soils, vegetation canopies, and built-up urban areas (producing characteristically low or negative MNDWI values). The MNDWI spectral index is formally defined:

$$\text{MNDWI} = \frac{\rho_{\text{Green}} - \rho_{\text{SWIR}}}{\rho_{\text{Green}} + \rho_{\text{SWIR}}} \quad (2)$$

where ρ_{Green} and ρ_{SWIR} represent surface reflectance values in green visible and shortwave infrared spectral bands respectively. A fixed threshold value of zero was systematically applied to generate binary water/non-water classification masks from continuous MNDWI index values.

Following initial thresholding operations, connected component analysis algorithms were systematically applied to identify spatially contiguous water body regions. Only the three largest spatially connected components per image were systematically retained, thereby eliminating isolated water pixels corresponding to small ponds, minor drainage channels, and ephemeral water features unrelated to primary Padma River channel. This connected component filtering substantially improves signal quality by removing spurious detections. All processed images were subsequently standardized via bilinear resampling to consistent 256×256 pixel spatial dimensions suitable for deep learning model input tensor specifications.

E. Comprehensive Gap-Filling Framework and Performance Benchmark

To systematically address inevitable data gap patterns arising from cloud contamination and sensor operational interruptions, a comprehensive gap-filling performance benchmark evaluation was meticulously conducted on yearly water mask temporal series (38 annual observations spanning 1988–2025). To simulate realistic missing-data conditions representative of operational scenarios, exactly 20% of temporal series observations—specifically seven randomly selected years (2002, 2005, 2010, 2015, 2019, 2020, 2024)—were systematically withheld as held-out test dataset, with all remaining observations utilized as ground truth reference for quantitative evaluation.

Seven distinct gap-filling strategic approaches underwent systematic comparative evaluation:

- 1) **Mean Temporal Composite:** Baseline reference method estimating missing yearly masks through simple arithmetic mean aggregation of temporally adjacent available observations, providing naive yet computationally efficient and robust spatial reconstructions.
- 2) **Median Temporal Composite:** Alternative baseline reference method substituting median statistical aggregation for mean, providing enhanced robustness against outlier observations while maintaining computational efficiency.
- 3) **Linear Temporal Interpolation:** Per-pixel linear interpolation computed between nearest available temporal neighbouring observations, yielding smooth temporal transition characteristics but exhibiting limited structural preservation capabilities for complex spatial features.
- 4) **Cubic Spline Temporal Interpolation:** Per-pixel cubic spline interpolation applied across temporal dimension, enabling smooth nonlinear pixel trajectory modelling with continuous first and second derivatives providing enhanced temporal smoothness relative to linear interpolation.
- 5) **Signed Distance Field (SDF) Interpolation:** Geometrically motivated approach wherein binary water masks are first transformed into signed distance field (SDF) representations defined as

$$\text{SDF}(\mathbf{x}) = d(\mathbf{x}, W^c) - d(\mathbf{x}, W), \quad (3)$$

where

$$d(\mathbf{x}, A) = \min_{\mathbf{y} \in A} \|\mathbf{x} - \mathbf{y}\|, \quad (4)$$

W denotes the water region, W^c represents its complement, $d(\mathbf{x}, A)$ is the Euclidean distance from location $\mathbf{x}$ to the set A , and $\|\cdot\|$ denotes the Euclidean norm. Positive SDF values correspond to pixels inside the water region, whereas negative values correspond to pixels outside the water region. Temporal interpolation is performed in the continuous SDF domain, followed by thresholding at the zero level set to recover binary

water masks. This approach preserves boundary geometry and topological consistency more effectively than direct interpolation of binary masks.

- 6) **Inverse-Distance Weighted Temporal Fusion:** Missing mask reconstructions are computed through temporally weighted averaging of all available yearly observations, with weights inversely proportional to their temporal distance from the target year:

$$M_t = \frac{\sum_{i \in A} w_i M_i}{\sum_{i \in A} w_i}, \quad w_i = \frac{1}{|t - i|} \quad (5)$$

where M_t denotes the reconstructed mask at target year t , M_i represents an available observation at year i , and A denotes the set of all available observations. Consequently, observations temporally closer to the target year exert greater influence on the reconstruction than more distant observations.

- 7) **Bidirectional Convolutional LSTM (BiConvLSTM):** Deep learning-based approach wherein bidirectional convolutional LSTM architecture is systematically trained on 128×128 pixel spatiotemporal patches extracted randomly from full temporal series. Training protocol incorporates 20% random temporal masking simulation to replicate missing observation patterns. The model architecture learns both forward and backward temporal dependency patterns and was optimized using binary cross-entropy loss over maximum 100 training epochs with early stopping (patience = 15). Adam optimizer ($\eta = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$) with batch size 16 was employed.

Quantitative performance results are comprehensively summarized in Table II. BiConvLSTM achieved best overall performance across all evaluation metrics (IoU = 0.7366, Dice = 0.8477, Recall = 0.8511), conclusively confirming its superior capability to capture extraordinarily complex spatiotemporal dynamic patterns. Among classical non-learning methods, SDF interpolation performed strongest (IoU = 0.7290, Dice = 0.8429), closely approaching deep learning performance levels while maintaining excellent structural consistency of water body boundary characteristics. Standard mean/median temporal composites and linear interpolation exhibited comparable baseline performance (IoU $\approx$ 0.655), while spline interpolation and weighted temporal fusion strategies offered modest incremental improvements.

V. COMPREHENSIVE METHODOLOGY

This section provides exhaustive technical specification of all methodological components including sequence generation protocols, temporal splitting strategies, comprehensive model architecture specifications with mathematical formulations, loss function definitions, evaluation metric specifications, training hyperparameter configurations, and risk zone mapping computational framework.

TABLE II
COMPREHENSIVE PERFORMANCE COMPARISON OF SEVEN GAP-FILLING
METHODS EVALUATED ON SEVEN HELD-OUT YEARLY OBSERVATIONS

Method	IoU	Dice	Precision	Recall
Mean Composite	0.6573	0.7926	0.8535	0.7403
Median Composite	0.6573	0.7926	0.8535	0.7403
Linear Interp.	0.6548	0.7910	0.8913	0.7193
Spline Interp.	0.6733	0.8042	0.8082	0.8012
SDF	0.7290	0.8429	0.8741	0.8161
Weighted Temporal	0.6724	0.8035	0.8306	0.7790
BiConvLSTM	0.7366	0.8477	0.8457	0.8511

A. Supervised Sequence Generation Strategy

To systematically model temporal morphological evolution as supervised learning problem, preprocessed binary water masks were transformed into structured input–output sequence pairs using sliding temporal window approach. For each investigated window length $L \in \{4, 5, 6\}$ (evaluated for yearly and quarterly temporal resolutions) or $L \in \{6, 9\}$ (evaluated for bi-monthly temporal resolution), exactly L consecutive temporal observations serve as input feature tensor and immediate chronologically next time step serves as prediction target:

$$\mathbf{X}_i = (X_i, X_{i+1}, \dots, X_{i+L-1}), \quad y_i = X_{i+L} \quad (6)$$

where $X_t \in \{0, 1\}^{H \times W}$ represents binary water mask at time t with spatial dimensions $H \times W = 256 \times 256$ pixels. Temporal stride of one year (or one bi-monthly period for higher-frequency data) generates systematically overlapping sequences, maximizing utilization of available but limited temporal data. This multi-sequence evaluation protocol enables systematic identification of optimal temporal context window length for each distinct architecture and temporal resolution combination.

B. Rigorous Temporal Splitting Strategy

A meticulously rigorous temporal split strategy prevents any data leakage between training and evaluation datasets. Two experimental evaluation setups are formally defined:

- **Setup 1 (Long-Range Forecasting):** Training incorporates all available data up to and including calendar year 2015; testing evaluates forecast performance on 2016–2025 inclusive (providing 10-year forecasting horizon for yearly data, or equivalent temporal span for quarterly/bi-monthly aggregations).
- **Setup 2 (Medium-Range Forecasting):** Training incorporates all available data up to and including calendar year 2020; testing evaluates forecast performance on 2021–2025 inclusive (providing 5-year forecasting horizon for yearly data, or equivalent temporal span for quarterly/bi-monthly aggregations).

For quarterly temporal resolution data, temporal cutoffs correspond to Q4-2015 and Q4-2020 respectively. For bi-monthly resolution, cutoffs correspond to P6-2015 and P6-2020 (November–December 2015 and 2020). Within each

setup configuration, final 15% of training sequences (ordered chronologically) are systematically allocated to validation dataset for hyperparameter tuning and early stopping criteria. Critically, absolutely zero temporal overlap exists between training, validation, and testing datasets, ensuring evaluation integrity.

C. Comprehensive Model Architecture Specifications

Five distinct spatiotemporal deep learning architectures underwent systematic comparative evaluation, selected to comprehensively span the architectural spectrum from recurrent convolutional hybrids to pure attention-based transformer models.

1) **Standalone ConvLSTM Architecture:** The ConvLSTM architecture [8] fundamentally replaces standard LSTM matrix multiplication operations with spatial convolution operations throughout all gate mechanisms, thereby enabling intrinsic joint spatial–temporal processing without requiring separate encoder structure. The complete network architecture comprises:

- 1) **Initial Spatial Feature Extraction:** Two successive spatial Conv2D layers with 32 and 64 convolutional filters (kernel size 3×3 , ReLU activation, same padding) extract initial low-level spatial features.
- 2) **Spatial Downsampling:** Batch normalization followed by max-pooling (2×2 window, stride 2) and spatial dropout (rate = 0.2) applied after each Conv2D layer.
- 3) **Spatiotemporal Processing:** Two stacked ConvLSTM2D layers with 128 and 64 filters respectively (kernel size 3×3). The first ConvLSTM2D layer processes full input sequence with `return_sequences=True`; the second layer aggregates temporal information with `return_sequences=False`, outputting single feature map.
- 4) **Upsampling Decoder:** Symmetric decoder comprising two upsampling blocks (UpSampling2D with factor 2 followed by Conv2D with 64 and 32 filters), batch normalization, and final 1×1 Conv2D with sigmoid activation producing prediction probability map.

2) **U-Net with LSTM Architecture:** The U-Net+LSTM hybrid model [28], [29] processes input sequence through TimeDistributed encoder extracting multi-scale spatial features independently for each temporal frame. Complete architecture specification:

- 1) **TimeDistributed Encoder:** Three successive encoder blocks each comprising: Conv2D layer (filters: 64, 128, 256 for blocks 1–3; kernel 3×3 , ReLU), batch normalization, second Conv2D with identical specifications, max-pooling (2×2), spatial dropout (rate = 0.2). TimeDistributed wrapper ensures identical processing applied to each input frame independently.
- 2) **Bottleneck Temporal Fusion:** LSTM module (128 units) at encoder bottleneck models temporal dependencies across sequence, producing single temporally-fused feature volume.

3) **Symmetric Decoder with Skip Connections:** Three decoder blocks each comprising: UpSampling2D (factor 2), concatenation with corresponding encoder skip connection feature map, two Conv2D layers (filters decreasing: 256, 128, 64), batch normalization. Final 1×1 Conv2D with sigmoid activation produces prediction.

3) **Attention U-Net with ConvLSTM Architecture:** The Attention U-Net+ConvLSTM model extends U-Net+LSTM through incorporating sophisticated attention gate mechanisms in each skip connection pathway [22], [23]. For encoder feature map $\mathbf{x}^l \in \mathbb{R}^{H_l \times W_l \times C_l}$ at encoder level l and gating signal $\mathbf{g} \in \mathbb{R}^{H_l \times W_l \times C_g}$ derived from corresponding decoder level, attention coefficient computation proceeds via:

$$\mathbf{q}^l = W_x * \mathbf{x}^l + W_g * \mathbf{g} + b_g \quad (7)$$

$$\alpha^l = \sigma(W_\psi * \text{ReLU}(\mathbf{q}^l) + b_\psi) \quad (8)$$

where $*$ denotes 1×1 convolution operation, W_x , W_g , W_ψ represent learnable convolutional kernel parameters, b_g , b_ψ are bias terms, and σ denotes sigmoid activation function producing spatial attention coefficients $\alpha^l \in [0, 1]^{H_l \times W_l}$. The skip connection output becomes element-wise product:

$$\tilde{\mathbf{x}}^l = \alpha^l \odot \mathbf{x}^l \quad (9)$$

thereby suppressing background regions (low α values) and amplifying morphologically active zones including channel margins and active erosion fronts (high α values). Critically, the bottleneck employs ConvLSTM2D rather than standard LSTM, preserving spatial locality during temporal fusion operations. Fig. ?? illustrates complete architecture.

4) **Swin Spatio-Temporal Transformer Architecture:** The Swin Spatio-Temporal Transformer [31] applies shared hierarchical Swin Transformer encoder to each input frame independently. Complete architecture specification:

Shared Swin Encoder: Each stage applies window-based multi-head self-attention with shifted window partitioning strategy (W-MSA and SW-MSA) followed by patch merging for progressive spatial downsampling. Three encoder stages produce feature maps at spatial resolutions $H/4 \times W/4$, $H/8 \times W/8$, and $H/16 \times W/16$ with embedding dimensions 64, 128, and 256 respectively. Self-attention within each window $\mathcal{W}$ of size $M \times M$ is computed as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{SoftMax} \left(\frac{\mathbf{QK}^T}{\sqrt{d}} + \mathbf{B} \right) \mathbf{V} \quad (10)$$

where $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ represent query, key, value projections, d is head dimension, and $\mathbf{B}$ encodes relative position biases. Shifted windowing across successive layers enables cross-window connections.

Temporal Fusion Module: Features at finest encoder resolution across all T input frames are stacked forming spatiotemporal tensor $\mathbf{F} \in \mathbb{R}^{T \times (H/4) \times (W/4) \times 64}$. Temporal fusion comprises:

1) 1D temporal convolution (kernel size 3, GELU activation) applied across temporal dimension at each spatial location.

2) Two cross-temporal attention layers (4 heads each) enabling each spatial location to attend across all temporal frames.

3) Softmax-based temporal aggregation producing single fused feature map $\mathbf{F} \in \mathbb{R}^{(H/4) \times (W/4) \times 64}$.

U-Net Style Decoder: Symmetric decoder with skip connections from encoder stages 1 and 2 progressively upsamples to full resolution.

5) **Vision Transformer-Based Spatio-Temporal Model:** The ViT-based model [32], [33] applies shared Vision Transformer encoder to each frame independently. Complete specification:

Patch Embedding: Each frame is partitioned into non-overlapping 16×16 patches yielding $(256/16)^2 = 256$ patch tokens. Linear projection maps each flattened patch to 128-dimensional embedding space:

$$\mathbf{z}_p = \mathbf{E} \cdot \text{vec}(\mathbf{x}_p) + \mathbf{e}_{\text{pos}} \quad (11)$$

where $\mathbf{E} \in \mathbb{R}^{128 \times (16 \times 16)}$ represents learnable projection matrix, $\mathbf{e}_{\text{pos}}$ encodes spatial position.

Spatial Transformer Blocks: Two successive transformer blocks (multi-head self-attention with 4 heads, MLP with hidden dimension 256, LayerNorm, residual connections, dropout 0.1) process patch token sequence.

Temporal Aggregation: Global average pooling aggregates all patch tokens yielding per-frame feature vector $\mathbf{f}_t \in \mathbb{R}^{128}$. Frame features are stacked, augmented with learned temporal positional embeddings, and processed by two temporal transformer layers (4 heads, MLP dimension 256) modeling inter-frame dependencies.

Convolutional Decoder: Aggregated temporal feature is spatially projected to $16 \times 16 \times 128$ feature map, then decoded through four transposed convolution blocks progressively up-sampling to full resolution. Fig. ?? illustrates architecture.

D. Loss Function Specifications

All models underwent training optimization using combined Dice and Focal loss formulation:

$$\mathcal{L}_{\text{Total}} = \lambda_1 \mathcal{L}_{\text{Dice}} + \lambda_2 \mathcal{L}_{\text{Focal}} \quad (12)$$

where weighting coefficients $\lambda_1 = \lambda_2 = 0.5$ provide equal contribution.

Dice Loss directly optimizes Dice overlap metric and proves particularly effective for class-imbalanced segmentation where water pixels constitute spatial minority:

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{2 \sum_i p_i g_i + \varepsilon}{\sum_i p_i + \sum_i g_i + \varepsilon} \quad (13)$$

where p_i represents predicted probability for pixel i , $g_i \in \{0, 1\}$ represents ground truth label, and $\varepsilon = 1$ provides numerical stability smoothing.

Focal Loss systematically down-weights easily-classified background examples to concentrate gradient updates on challenging boundary pixels [34]:

$$\mathcal{L}_{\text{Focal}} = -\frac{1}{N} \sum_{i=1}^N \alpha_i (1 - p_{t_i})^\gamma \log(p_{t_i}) \quad (14)$$

where $p_{t_i} = p_i$ if $g_i = 1$ else $1 - p_i$, class weighting $\alpha_i = 0.75$ for positive class, focusing parameter $\gamma = 2.0$, and N represents total pixel count. The combined loss strategically leverages complementary strengths: Dice loss ensures global overlap accuracy; Focal loss provides precise boundary-level supervision concentrating learning on morphologically critical transition zones.

E. Comprehensive Evaluation Metrics

Model performance underwent comprehensive assessment using five complementary quantitative metrics evaluated at binarization threshold of 0.5:

Intersection over Union (IoU):

$$\text{IoU} = \frac{|A \cap B| + \varepsilon}{|A \cup B| + \varepsilon} = \frac{\text{TP} + \varepsilon}{\text{TP} + \text{FP} + \text{FN} + \varepsilon} \quad (15)$$

Dice Coefficient (F1-Score):

$$\text{Dice} = \frac{2|A \cap B| + \varepsilon}{|A| + |B| + \varepsilon} = \frac{2 \text{TP} + \varepsilon}{2 \text{TP} + \text{FP} + \text{FN} + \varepsilon} \quad (16)$$

Precision:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (17)$$

Recall (Sensitivity):

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (18)$$

where TP, FP, FN represent true positives, false positives, and false negatives respectively, $\varepsilon = 10^{-7}$ provides numerical stability.

Additionally, **absolute area difference** quantifies spatial calibration independently of overlap metrics:

$$\Delta A = A_{\text{pred}} - A_{\text{true}} = (N_{\text{pred}} - N_{\text{true}}) \times A_{\text{pixel}} \quad (19)$$

where N represents pixel counts, $A_{\text{pixel}} = 0.0036 \text{ km}^2$ at 60 m resolution. A persistence baseline (predicting last observed frame) provides reference lower bound.

F. Training Configuration and Hyperparameters

CNN-LSTM Hybrid Models (ConvLSTM, U-Net+LSTM, Attention U-Net+ConvLSTM): All three architectures employed identical training configuration for fair comparison. Optimization utilized Adam algorithm [39] with learning rate $\eta = 1 \times 10^{-4}$, momentum parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, and gradient clipping via `clipnorm = 1.0` preventing exploding gradient instability. Batch size of 4 sequences balanced GPU memory constraints against gradient estimate quality. Training proceeded for maximum 200 epochs with early stopping monitoring validation loss (patience = 20 epochs, restore best weights). ReduceLROnPlateau callback dynamically adjusted learning rate upon validation loss plateau detection (patience = 7 epochs, reduction factor = 0.5, minimum $\eta_{\min} = 1 \times 10^{-7}$). Model checkpointing systematically saved best-performing checkpoint based on validation loss criterion.

Swin Spatio-Temporal Transformer: Reduced batch size of 2 sequences accommodated substantially elevated GPU

memory requirements (patch size = 4, embedding dimension = 64, window size = 8, hierarchical depths [2, 2, 2], attention heads [2, 4, 8], MLP ratio = 2.0, spatial dropout = 0.1). Temporal fusion module configured with 2 temporal transformer layers, 4 attention heads, MLP dimension 256, dropout 0.1. Decoder filter progression: [128, 64, 32, 16]. Remaining training hyperparameters identical to CNN-LSTM models.

ViT-Based Spatio-Temporal Model: Batch size 2 (patch size = 16, embedding dimension = 128, 2 spatial transformer layers, 2 temporal transformer layers, 4 attention heads each, MLP dimension 256, dropout 0.1). Remaining hyperparameters identical.

All implementations employed fixed random seeds (NumPy: 42, TensorFlow: 42) ensuring reproducibility, and utilized identical data split configurations enabling rigorous controlled comparison.

G. Risk Zone Mapping Computational Framework

Following generation of iterative autoregressive forecasts spanning 2026–2045 horizon, comprehensive post-processing framework identifies erosion and accretion risk zones at pixel-level spatial resolution. **Erosion** is rigorously defined by land-to-water transition between consecutive forecast years ($L_t = 0 \wedge L_{t+1} = 1$); **accretion** defined by reverse water-to-land transition. Frequency maps quantify temporal persistence:

$$f_{\text{erosion}}(\mathbf{x}) = \frac{1}{T-1} \sum_{t=1}^{T-1} \mathbb{I}(L_t(\mathbf{x}) = 0 \wedge L_{t+1}(\mathbf{x}) = 1) \quad (20)$$

$$f_{\text{accretion}}(\mathbf{x}) = \frac{1}{T-1} \sum_{t=1}^{T-1} \mathbb{I}(L_t(\mathbf{x}) = 1 \wedge L_{t+1}(\mathbf{x}) = 0) \quad (21)$$

where $\mathbf{x}$ denotes spatial pixel location, T represents total forecast years, $\mathbb{I}(\cdot)$ is indicator function.

A **hydrodynamic stability index** quantifies pixel-level classification uncertainty via water occurrence frequency:

$$\mathcal{I}(\mathbf{x}) = 1 - |2F_w(\mathbf{x}) - 1|, \quad F_w(\mathbf{x}) = \frac{1}{T} \sum_{t=1}^T L_t(\mathbf{x}) \quad (22)$$

where $F_w \in [0, 1]$ represents fraction of forecast period classified as water. Values $\mathcal{I} \approx 1$ indicate persistent land–water oscillation (maximum instability); $\mathcal{I} \approx 0$ indicates stable land or water classification (maximum stability).

Composite risk representation integrates erosion frequency, accretion frequency, and instability index highlighting regions simultaneously dynamic and uncertain. Physical area estimates employ:

$$A_{\text{erosion/accretion}} = N_{\text{pixels}} \times A_{\text{pixel}} \quad (23)$$

where $A_{\text{pixel}} = (60 \text{ m})^2 = 0.0036 \text{ km}^2$ at established 60 m spatial resolution.

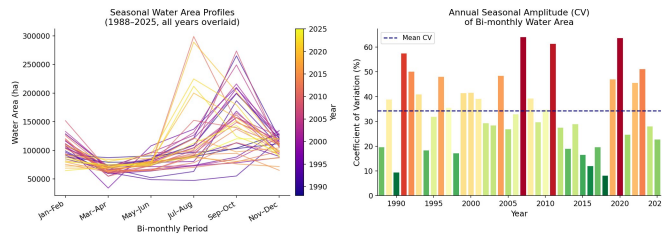


Fig. 3. Seasonal water area dynamics of the Padma River (1988–2025). Left: bi-monthly water area profiles for all years, coloured by year. Right: annual coefficient of variation (CV) of bi-monthly water area, representing inter-annual differences in flood-pulse intensity and hydrological variability.

VI. SPATIOTEMPORAL STATISTICAL ANALYSIS

This section presents comprehensive quantitative spatiotemporal statistical characterization of 38 years of Padma River morphodynamic behaviour (1988–2025) including long-term areal dynamics, seasonal variability patterns, erosion–accretion mass budget calculations, and channel centreline migration rate quantification.

A. Long-Term Areal Dynamics and Trend Analysis

The Padma River exhibited substantial inter-annual variability in water-covered areal extent throughout 38-year observation period. Annual water-covered area ranged from absolute minimum 540.4 km² (calendar year 1995, hydrologically dry year characterized by below-average monsoon precipitation) to absolute maximum 1,229.3 km² (calendar year 1999, catastrophic monsoon flood corresponding to major regional disaster event), with long-term temporal mean of 664.5 ± 112.8 km² (arithmetic mean $\pm$ one standard deviation). The exceptionally large channel expansion manifest in 1999 corresponds temporally to the catastrophic July–August 1998 monsoon flood event representing one of most severe recorded flood disasters in Bangladesh’s documented history.

Formal non-parametric Mann–Kendall trend test [35], [36] indicated absence of statistically significant monotonic trend at $\alpha = 0.05$ significance level (Kendall’s $\tau = -0.149$, two-tailed $p = 0.187$), and Theil–Sen robust linear estimator [37], [38] yielded weak negative slope magnitude of -1.41 km² yr⁻¹. These complementary statistical results conclusively confirm that despite pronounced inter-annual fluctuations driven by monsoon variability, the long-term temporal mean river extent has remained broadly stable across the 38-year observation period, with no compelling statistical evidence supporting sustained systematic expansion or contraction trends.

B. Seasonal Variability and Intra-Annual Dynamics

Seasonal variability magnitude, systematically assessed via coefficient of variation (CV) computed from bi-monthly water area time series, demonstrated pronounced inter-annual differences (Fig. 3). Annual CV magnitude ranged from minimum 8.2% (calendar year 2018, most hydrologically stable year exhibiting minimal seasonal fluctuation) to maximum 64.1% (calendar year 2007, most hydrologically variable year), with



Fig. 4. Annual and decadal erosion and accretion of the Padma River channel (1988–2025). Top: annual gross erosion (red), gross accretion (blue), and net channel change (black line). Bottom: decadal grouped summary illustrating long-term morphodynamic phases.

study-period arithmetic mean CV of $34.3 \pm 13.7\%$. Calendar years exhibiting elevated CV values—particularly 2007, 2011, and 2020—were systematically associated with extreme monsoon flooding events during which bi-monthly peak water extent exceeded 2,700 km² during maximum inundation periods. Conversely, years characterized by low CV values indicate relatively stable seasonal water extent trajectories and comparatively weak flood-pulse variability magnitude.

The quarterly heat-map visualization (Fig. 3, bottom panel) comprehensively confirms persistent and highly consistent hydrological seasonal pattern maintained across all 38 observation years: Q4 (October–December, post-monsoon recession period) consistently exhibits largest mean water extent while Q2 (April–June, pre-monsoon low-flow period) systematically records minimum extent. This pronounced and remarkably stable seasonal structure directly reflects dominant climatological control exerted by South Asian monsoon meteorological system on Padma River hydrodynamics.

C. Comprehensive Erosion and Accretion Dynamics

Annual erosion and accretion rate magnitudes demonstrated exceptionally high inter-annual variability reflecting pronounced sensitivity to individual monsoon flood events (Fig. 4). The largest documented single-year erosion event occurred during 1998–1999 interval (649.1 km² land surface converted to water classification; net channel expansion 567.7 km²), corresponding temporally to catastrophic July–August 1998 monsoon flood representing most extreme channel widening event recorded throughout entire 38-year study period. Conversely, the immediately following year (1999–2000 interval) recorded largest net accretion magnitude (-503.5 km²), indicating substantial rapid channel contraction during post-flood geomorphic recovery and adjustment phase. This dramatic reversal pattern highlights pronounced

TABLE III

DECADAL EROSION, ACCRETION, AND NET CHANNEL CHANGE
SUMMARY FOR PADMA RIVER (1988–2025). POSITIVE NET CHANGE
VALUES INDICATE CHANNEL EXPANSION (EROSION-DOMINATED REGIME);
NEGATIVE VALUES INDICATE CHANNEL CONTRACTION
(ACCRETION-DOMINATED REGIME).

Decadal Period	Erosion (km ²)	Accretion (km ²)	Net Change (km ²)
1988–1998	367.6	434.2	−66.7
1998–2008	382.3	355.9	+26.3
2008–2018	262.8	375.8	−113.0
2018–2025	264.6	264.3	+0.3
Total	5170.2	5323.3	−153.0

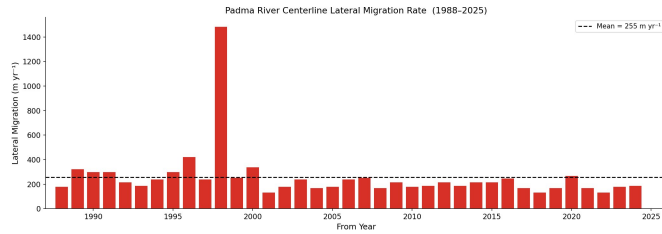


Fig. 5. Annual lateral migration rate of the Padma River centreline (1988–2025). Red bars indicate yearly migration rates, while the dashed line marks the long-term mean of 255.4 m yr^{−1}. The extreme peak in 1998–1999 corresponds to the catastrophic flood-driven channel adjustment.

recovery dynamics characterizing post-extreme-event channel behaviour.

Table III provides systematic quantitative summary of decadal erosion–accretion mass budgets.

Across complete 38-year study period, cumulative gross erosion reached 5,170.2 km² (representing multiple complete channel reworking cycles) while cumulative gross accretion totalled 5,323.3 km², yielding overall net accretion magnitude of 153.0 km² indicating slight long-term channel contraction tendency. Decadal analysis systematically identifies four temporally distinct morphodynamic regime phases: (i) 1988–1998 exhibited near-balanced erosion–accretion dynamics (−66.7 km² net accretion) characteristic of lateral channel migration without sustained widening; (ii) 1998–2008 shifted toward slight net erosion tendency (+26.3 km²) dominated by catastrophic 1998–1999 flood event and subsequent multi-year adjustment; (iii) 2008–2018 manifested strongest net accretion (−113.0 km²) suggesting significant channel contraction and partial geomorphic stabilization; (iv) 2018–2025 approached near-equilibrium conditions with negligible net change (+0.3 km²) indicating contemporary period of relative morphodynamic stability.

D. Channel Centreline Migration Rate Quantification

Channel centreline lateral migration rates exhibited pronounced temporal variability reflecting alternating periods of geomorphic instability and relative stabilization (Fig. 5).

TABLE IV

PERFORMANCE COMPARISON OF ALL ARCHITECTURES ON YEARLY DATA

Model	Setup	IoU	Dice	P	R	ΔA
ConvLSTM	S1	0.6218	0.7665	0.7212	0.8207	76.69
ConvLSTM	S2	0.6278	0.7711	0.7139	0.8432	103.26
U-Net+LSTM	S1	0.6964	0.8208	0.7876	0.8595	50.25
U-Net+LSTM	S2	0.6986	0.8223	0.7905	0.8592	48.61
Att.-U-Net+CLSTM	S1	0.7005	0.8236	0.7807	0.8747	66.54
Att.-U-Net+CLSTM	S2	0.6963	0.8205	0.7934	0.8523	41.05
ViT	S1	0.3613	0.5302	0.4352	0.6806	316.06
ViT	S2	0.3831	0.5526	0.5197	0.5907	78.69
Swin	S1	0.6080	0.7560	0.7364	0.7780	31.64
Swin	S2	0.5912	0.7431	0.6849	0.8122	96.23

Arithmetic mean annual migration rate across full study period was 255.4 m yr^{−1}, with individual year magnitudes ranging from minimum 134.2 m yr^{−1} (calendar years 2001–2002 and 2022–2023, representing temporary stabilization phases) to catastrophic maximum 1,485.5 m yr^{−1} during 1998–1999 interval (corresponding to extreme flood-driven channel adjustment event). The median migration rate (216.3 m yr^{−1}) was substantially below arithmetic mean, conclusively confirming that small number of extreme flood years strongly inflated long-term average metrics while typical inter-annual behaviour exhibits more moderate migration magnitudes. The 1990s decade recorded highest average migration rates (mean 348.7 m yr^{−1}), definitively designating that decade as period of maximum geomorphic instability within 38-year observational record. Cumulative centreline displacement over complete study period exceeded 9 km total lateral adjustment, demonstrating sustained long-term lateral migration and continuous comprehensive reworking of active river corridor.

Attention U-Net+ConvLSTM achieved highest IoU (0.7005) and Dice (0.8236) in Setup 1, establishing it as optimal architecture for yearly prediction tasks. U-Net+LSTM demonstrated competitive performance with slightly lower overlap metrics but superior area calibration. ConvLSTM and Swin Transformer exhibited moderate performance, while Vision Transformer significantly underperformed all alternative architectures.

1) *Detailed Year-Wise Analysis:* Tables V and VI present comprehensive year-by-year breakdown for best-performing Attention U-Net+ConvLSTM architecture.

Performance remained remarkably stable across 10-year horizon with IoU ranging 0.6569–0.7359 (standard deviation 0.0273). Higher Recall (mean 0.8747) relative to Precision (0.7807) indicates systematic tendency toward over-prediction, consistent with positive area differences in most years. The 5-year setup achieved lower area differences (mean 41.05 km² vs 66.54 km²), confirming shorter horizons reduce temporal uncertainty.

TABLE V
DETAILED YEAR-WISE PREDICTION RESULTS: ATTENTION
U-NET+CONVLSTM, 10-YEAR FORECASTING SETUP (2016–2025)

Year	IoU	Dice	Prec.	Rec.	ΔA (km ²)
2016	0.7357	0.8477	0.8655	0.8307	−24.96
2017	0.6975	0.8218	0.7396	0.9246	130.72
2018	0.6742	0.8054	0.7273	0.9023	122.18
2019	0.7051	0.8270	0.7858	0.8729	58.46
2020	0.7055	0.8273	0.7924	0.8654	51.24
2021	0.6569	0.7929	0.7706	0.8165	34.81
2022	0.7359	0.8479	0.8407	0.8552	10.84
2023	0.7081	0.8291	0.7618	0.9095	107.73
2024	0.7046	0.8267	0.7806	0.8786	70.94
2025	0.6814	0.8105	0.7430	0.8915	103.46
Mean	0.7005	0.8236	0.7807	0.8747	66.54
Std	0.0273	0.0179	0.0440	0.0325	56.23

TABLE VI
DETAILED YEAR-WISE PREDICTION RESULTS: ATTENTION
U-NET+CONVLSTM, 5-YEAR FORECASTING SETUP (2021–2025)

Year	IoU	Dice	Prec.	Rec.	ΔA (km ²)
2021	0.6384	0.7793	0.7797	0.7788	−0.66
2022	0.7270	0.8419	0.8546	0.8296	−18.39
2023	0.7143	0.8333	0.7745	0.9018	91.31
2024	0.7000	0.8235	0.7950	0.8542	42.04
2025	0.7017	0.8247	0.7631	0.8972	90.98
Mean	0.6963	0.8205	0.7934	0.8523	41.05
Std	0.0341	0.0244	0.0334	0.0511	50.98

TABLE VII
PERFORMANCE COMPARISON OF ALL ARCHITECTURES ON QUARTERLY
DATA

Model	Setup	IoU	Dice	P	R	$ \Delta A $
ConvLSTM	S1	0.6114	0.7572	0.7433	0.7918	166.66
ConvLSTM	S2	0.6134	0.7586	0.7115	0.8348	159.91
U-Net+LSTM	S1	0.6908	0.8095	0.8162	0.8318	188.03
U-Net+LSTM	S2	0.7008	0.8191	0.8257	0.8431	163.76
Att.-U-Net+CLSTM	S1	0.6956	0.8139	0.8095	0.8443	184.80
Att.-U-Net+CLSTM	S2	0.6972	0.8170	0.8054	0.8511	137.98
ViViT	S1	0.3648	0.5316	0.4788	0.6347	340.08
ViViT	S2	0.3822	0.5496	0.4606	0.7281	447.09
Swin	S1	0.5865	0.7370	0.7012	0.7981	190.79
Swin	S2	0.5882	0.7392	0.7004	0.8024	160.45

E. Quarterly Prediction Results: Complete Architecture Comparison

Attention U-Net+ConvLSTM and U-Net+LSTM exhibited closely competitive performance for quarterly prediction, with Attention model achieving marginally superior IoU in Setup 1

TABLE VIII
DETAILED QUARTERLY PREDICTION RESULTS – SETUP 1 (ATTENTION
U-NET+CONVLSTM, SEQLEN=8, TEST: 2017 Q2–2025 Q4)

Quarter	IoU	Dice	Prec.	Rec.	ΔA (km ²)
2017-Q2	0.9156	0.9559	0.9524	0.9595	5.31
2017-Q3	0.7556	0.8608	0.9543	0.7840	−171.18
2017-Q4	0.7778	0.8750	0.9127	0.8404	−73.25
2018-Q1	0.8315	0.9080	0.8833	0.9341	41.75
2018-Q2	0.9150	0.9556	0.9304	0.9823	38.33
2018-Q3	0.7056	0.8274	0.8341	0.8208	−12.15
2018-Q4	0.5594	0.7175	0.6733	0.7678	94.13
2019-Q1	0.8782	0.9351	0.8851	0.9911	71.73
2019-Q2	0.6949	0.8200	0.7197	0.9528	197.74
2019-Q3	0.5647	0.7218	0.9123	0.5971	−447.49
2019-Q4	0.6155	0.7620	0.6746	0.8754	241.01
2020-Q1	0.7440	0.8532	0.7610	0.9708	157.89
2020-Q2	0.9068	0.9511	0.9353	0.9675	20.12
2020-Q3	0.5061	0.6721	0.8699	0.5475	−565.15
2020-Q4	0.5661	0.7230	0.6572	0.8034	179.91
2021-Q1	0.7922	0.8840	0.8376	0.9360	72.49
2021-Q2	0.8128	0.8967	0.8156	0.9958	119.94
2021-Q3	0.4833	0.6517	0.8805	0.5173	−662.31
2021-Q4	0.5279	0.6910	0.5910	0.8316	320.34
2022-Q1	0.6335	0.7756	0.6908	0.8842	142.33
2022-Q2	0.6190	0.7647	0.9081	0.6604	−213.69
2022-Q3	0.5230	0.6868	0.8612	0.5711	−519.60
2022-Q4	0.6256	0.7697	0.7454	0.7956	63.38
2023-Q1	0.7641	0.8663	0.8414	0.8926	38.33
2023-Q2	0.7147	0.8336	0.7684	0.9109	107.41
2023-Q3	0.6263	0.7702	0.9154	0.6647	−306.30
2023-Q4	0.6325	0.7749	0.6465	0.9670	359.43
2024-Q1	0.6504	0.7882	0.6631	0.9713	215.20
2024-Q2	0.7352	0.8474	0.8906	0.8081	−52.76
2024-Q3	0.5501	0.7097	0.5527	0.9915	427.37
2024-Q4	0.6480	0.7864	0.7055	0.8882	174.97
2025-Q1	0.8124	0.8965	0.8238	0.9833	118.42
2025-Q2	0.9298	0.9636	0.9493	0.9783	18.22
2025-Q3	0.5993	0.7495	0.8262	0.6858	−170.04
2025-Q4	0.7280	0.8426	0.8655	0.8209	−48.20
Mean	0.6956	0.8139	0.8095	0.8443	−0.47
Std	0.1230	0.0851	0.1124	0.1275	244.68

(0.6956 vs 0.6908).

1) *Detailed Quarterly Results: Setup 1:* Table VIII presents complete quarter-by-quarter breakdown for Attention U-Net+ConvLSTM covering 2017 Q2–2025 Q4 test period.

Pronounced seasonal stratification is evident: Q2 (pre-monsoon) consistently achieves $\text{IoU} > 0.90$ in multiple years (2017, 2018, 2020, 2025), while Q3 (peak monsoon) exhibits lowest performance, frequently falling below 0.55. The signed mean area difference approaches zero (−0.47 km²), indicating excellent long-term calibration despite substantial within-year fluctuations driven by seasonal effects.

2) *Detailed Quarterly Results: Setup 2:* Table IX presents detailed results for shorter 5-year forecasting horizon (2022 Q4–2025 Q4).

Setup 2 results demonstrate consistent seasonal patterns with improved mean performance ($\text{IoU} = 0.6972$ vs 0.6956) attributable to shorter forecasting horizon reducing cumulative error propagation.

TABLE IX
DETAILED QUARTERLY PREDICTION RESULTS – SETUP 2 (ATTENTION
U-NET+CONVLSTM, SEQLen=10, TEST: 2022 Q4–2025 Q4)

Quarter	IoU	Dice	Prec.	Rec.	ΔA (km ²)
2022-Q4	0.6086	0.7567	0.7498	0.7637	17.46
2023-Q1	0.7485	0.8561	0.8325	0.8812	36.82
2023-Q2	0.7025	0.8253	0.7826	0.8729	66.80
2023-Q3	0.6164	0.7627	0.8820	0.6719	−266.44
2023-Q4	0.6406	0.7810	0.6768	0.9230	263.79
2024-Q1	0.6552	0.7917	0.6769	0.9533	189.02
2024-Q2	0.7804	0.8767	0.9655	0.8028	−96.03
2024-Q3	0.5669	0.7236	0.5703	0.9894	395.49
2024-Q4	0.6496	0.7876	0.7263	0.8601	124.49
2025-Q1	0.8740	0.9328	0.9362	0.9293	−4.55
2025-Q2	0.9407	0.9695	0.9583	0.9809	14.04
2025-Q3	0.5745	0.7297	0.8544	0.6368	−254.68
2025-Q4	0.7061	0.8278	0.8583	0.7994	−64.14
Mean	0.6972	0.8170	0.8054	0.8511	32.47
Std	0.1043	0.0725	0.1111	0.1004	188.53

TABLE X
PERFORMANCE COMPARISON OF ALL ARCHITECTURES ON BI-MONTHLY
DATA

Model	Setup	IoU	Dice	P	R	$ \Delta A $
ConvLSTM	S1	0.6347	0.7732	0.7941	0.7708	131.44
ConvLSTM	S2	0.6623	0.7956	0.7979	0.8062	119.11
U-Net+LSTM	S1	0.7439	0.8435	0.8658	0.8496	156.64
U-Net+LSTM	S2	0.7791	0.8701	0.8971	0.8691	160.78
Att.-U-Net+CLSTM	S1	0.7410	0.8406	0.8806	0.8293	157.45
Att.-U-Net+CLSTM	S2	0.7733	0.8662	0.9094	0.8459	147.45
Swin	S1	0.6333	0.7718	0.8015	0.7608	131.99
Swin	S2	0.6952	0.8180	0.8404	0.8086	114.45
ViT	S1	0.3805	0.5487	0.5461	0.5771	203.91
ViT	S2	0.3421	0.5079	0.4723	0.5711	243.07

F. Bi-Monthly Prediction Results: Complete Architecture Comparison

U-Net+LSTM emerged as best-performing architecture for bi-monthly prediction, achieving IoU = 0.7791 and Dice = 0.8701 in Setup 2. This architectural shift—from Attention U-Net+ConvLSTM dominance at yearly/quarterly resolutions to U-Net+LSTM leadership at bi-monthly—suggests spatial feature extraction quality becomes progressively more critical as temporal frequency increases.

G. Long-Term Morphological Forecasting (2026–2040)

Long-term morphological forecasting employed final Attention U-Net+ConvLSTM model configuration trained on complete 1987–2025 dataset with sequence length five. Training convergence occurred at epoch 107 (best validation loss = 0.2298, validation Dice = 0.8057, validation IoU = 0.7026). Reconstruction-based sanity check on known historical period 2016–2025 yielded mean IoU = 0.6543

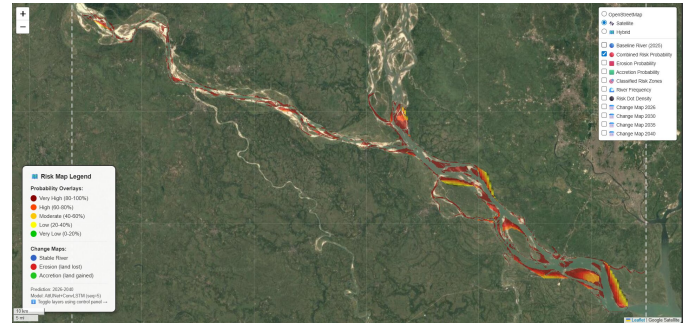


Fig. 6. Long-term combined morphological risk for 2026–2040. *Interactive web demonstration available here*

and mean Dice = 0.7907, confirming model’s capability to reproduce historical river extent patterns with acceptable consistency.

From 2025 baseline area of 517.6 km², predicted river extent increased progressively to 671.3 km² in 2026 and reached 807.6 km² by 2040, representing net expansion of 290.0 km² (+56.03% relative to baseline). Expansion rate exhibited non-uniform temporal pattern: largest incremental changes occurred during initial forecast years (2025–2026) with progressively declining annual increments, decreasing from 19.4 km² (2028–2029) to 4.6 km² (2039–2040). By terminal forecast year 2040, total projected morphological change relative to 2025 baseline comprised 35.5 km² erosion-type transitions (land→water), 325.5 km² accretion-type transitions (water→land in mask-based framework representing lateral expansion), and 482.1 km² stable river area maintaining consistent classification. The accretion-dominated pattern indicates future dynamics primarily associated with lateral expansion into adjacent floodplain areas rather than wholesale channel widening.

The comprehensive long-term risk cartography (Fig. 6) identifies spatially explicit high-probability morphological change zones concentrated systematically along active channel margins and proximal floodplain corridor segments, with very-high-risk designations indicating persistent instability and recurrent channel boundary adjustment patterns most operationally relevant for infrastructure protection planning and settlement management strategies.

H. Short-Term Bi-Monthly Forecasting (2026–2028)

Short-term high-frequency bi-monthly forecasting employed U-Net+ConvLSTM2D architecture with nine-period input sequences systematically enriched with sinusoidal and cosinusoidal seasonal encoding channels ($\sin(2\pi t/6)$, $\cos(2\pi t/6)$ where t represents bi-monthly period index), enabling model to explicitly distinguish intra-annual temporal periods without relying exclusively on spatial morphological content patterns. Training convergence occurred at epoch 39 (validation Dice = 0.8381, IoU = 0.7463). Reconstruction sanity check on recent historical period 2023 P1–2025 P6 yielded mean IoU = 0.684 ± 0.072 (mean $\pm$ std).

Over 18 forecast periods spanning 2026 P1–2028 P6, mean predicted area followed clear three-phase temporal structure: initial contraction phase to mean 902.7 km² in 2026 (below 934.1 km² baseline), reflecting dry-to-wet seasonal transition inherited from seed sequence; progressive rise to 984.0 km² in 2027; further increase to 1,020.1 km² in 2028, with terminal forecast period (2028 P6, November–December) reaching 1,054.4 km² (+12.88% above baseline). The seasonal cycle was faithfully reproduced with characteristic pattern: minimum extent at P3 (May–June pre-monsoon, 918.3 km²), maximum at P5 (September–October peak monsoon, 1,034.3 km²), seasonal amplitude 116.0 km², and mean monsoon area 53.4 km² (+5.62%) above non-monsoon mean. Final forecast period exhibited accretion-to-erosion ratio approximately 1.60:1, with 78.4% of baseline channel extent maintaining stable classification.

VII. DISCUSSION

Hybrid CNN-LSTM architectures demonstrated consistent superiority over pure transformer-based models across all datasets, temporal resolutions, and forecasting horizons. This finding aligns with Dehghani et al. [8], who reported ConvLSTM advantages for spatiotemporal geophysical forecasting, and with broader segmentation literature where U-Net variants remain competitive on limited training data [22], [23]. The underperformance of transformer-based architectures (ViT, ViViT, Swin Transformer) likely stems from data scarcity. The Padma River time series contains only 38 annual, 125 quarterly, or 156 bi-monthly observations, insufficient for training large-scale attention-based models without severe overfitting. Swin Transformer's advantage over standard ViT supports this, as hierarchical window-based attention imposes stronger spatial locality biases compensating for limited data. With expanding satellite archives and available pre-trained geospatial foundation models for transfer learning, transformer architectures are expected to become competitive with CNN-LSTM hybrids.

A clear resolution dependent pattern emerged from experiments. Attention U-Net+ConvLSTM dominated yearly and quarterly prediction, where attention mechanisms effectively focus on geomorphologically active zones. U-Net+LSTM performed best at bi-monthly resolution where fine boundary preservation proves critical and attention gates provide diminishing returns. This suggests operational systems should deploy distinct architectures for different temporal tasks rather than single universal models.

Seasonal forcing strongly stratified performance in quarterly and bi-monthly experiments. Q2 and P3 (April to June pre-monsoon) consistently achieved highest IoU values (occasionally exceeding 0.93), while Q3 and P4 (July to September peak monsoon) exhibited lowest performance (below 0.50). This reflects the challenge of forecasting peak monsoon extents when channels expand dramatically, adopt irregular planforms, and merge with inundated floodplains. Mean area differences approached zero for Setup 1 quarterly predictions (negative 0.47 km²), confirming systematic biases in opposite seasons

approximately cancel annually. Ensemble approaches combining separate models for dry and wet seasons could yield substantial improvements.

A critical finding for operational applications is that IoU and Dice metrics alone inadequately characterize model performance. Multiple configurations exhibited competitive overlap metrics but substantially different absolute area differences, indicating spatially correct predictions with dimensional mis-sizing. River management decisions including erosion risk estimation, agricultural land loss quantification, embankment design, and disaster resource allocation depend critically on accurate area estimation. We recommend reporting IoU, Dice, Precision, Recall, and absolute area difference as minimum evaluation standards, with area difference weighted heavily in model selection for hazard management.

Long-term forecasting (2026 to 2040) projects sustained lateral expansion with accretion dominating over erosion. This pattern aligns with observed 2008 to 2018 behavior (net accretion of 113.0 km²) and 2018 to 2025 near-equilibrium, suggesting models learned realistic long-term dynamics. The declining expansion rate after 2031 may reflect learned stabilization patterns, though cumulative forecasting errors could contribute. Consistent positive area bias in historical reconstruction (91.8 km²) suggests expansion magnitudes are upper bound estimates. However, spatial risk zone distributions identifying corridor segments with elevated change probability prove more reliable and constitute the primary operational product for disaster management.

A. Limitations and Future Research Directions

Several limitations warrant acknowledgement. First, 60 m Landsat resolution cannot resolve fine-scale erosion features, secondary channels, or detailed bank morphology; transitioning to 10 m Sentinel-2 data throughout the time series constitutes high priority. Second, absence of explicit hydrological forcing variables including river discharge, precipitation, and antecedent soil moisture represents important modelling gap, particularly for extreme flood years. Third, the framework does not incorporate geotechnical factors including soil type, bank height characteristics, or vegetation cover patterns that influence local erosion susceptibility. Fourth, all experiments were conducted on a single river system; rigorous cross-system validation on Jamuna and Meghna river systems is essential before confident broad generalization claims.

Priority directions for future research include: (i) systematic integration of 10 m Sentinel-2 and Sentinel-1 SAR data across the complete time series; (ii) geographic extension to Jamuna and Meghna river systems for cross-system validation; (iii) incorporation of hydrological forcing variables to enhance extreme event prediction; (iv) revisiting transformer architectures with pre-training on larger geospatial foundation model corpora; and (v) overlaying morphological risk cartography with population density, agricultural land use, and infrastructure datasets enabling fully integrated socioeconomic early warning systems for operational deployment.

VIII. CONCLUSION

This investigation presents a comprehensive framework for analyzing and forecasting morphological evolution of the Padma River by combining long-term satellite remote sensing with spatiotemporal deep learning.

A 38-year time series (1987-2025) of binary water masks was constructed from Landsat and Sentinel imagery via Google Earth Engine using MNDWI extraction. The proposed BiConvLSTM gap-filling model achieved superior performance (IoU=0.7366, Dice=0.8477) among seven evaluated methods. Spatiotemporal statistical analysis revealed: annual water area showed no significant long-term trend (Mann-Kendall $p=0.187$) but high inter-annual variability (coefficient of variation up to 64.15, 170.2 km² and accretion 5,323.3 km² across four decadal phases; mean centreline migration was 255.4 m yr⁻¹ with 1998 flood extreme of 1,485.5 m yr⁻¹).

Hybrid CNN-LSTM models consistently outperformed pure transformer architectures under limited data conditions. Attention U-Net+ConvLSTM achieved best yearly performance (IoU=0.7005, Dice=0.8236 over 10 years) and quarterly performance (IoU=0.6956). U-Net+LSTM led bi-monthly prediction (IoU=0.7791, Dice=0.8701). Long-term forecasting (2026 to 2040) projected 290 km² net expansion (56 identifying high-risk corridor segments. Bi-monthly forecasting (2026 to 2028) successfully captured monsoon seasonal cycles with 116 km² amplitude.

These results demonstrate that freely available satellite imagery combined with cloud-based processing and spatiotemporal deep learning provides a practical, scalable, and reproducible framework for riverbank hazard monitoring and multi-year forecasting. For millions inhabiting erosion-prone river corridors across Bangladesh and South Asia, analytical tools providing advance warning of high-risk zones can meaningfully enhance disaster preparedness, infrastructure planning, and community adaptation strategies.

Future research should: (i) integrate 10 m Sentinel-2 and Sentinel-1 SAR data across the complete time series; (ii) extend to Jamuna and Meghna river systems for cross-system validation; (iii) incorporate hydrological forcing variables for extreme event prediction; (iv) revisit transformer architectures with pre-training on larger geospatial foundation models; and (v) overlay morphological risk maps with population, agricultural land use, and infrastructure data to enable integrated socioeconomic early warning systems.

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